Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	29-09-2026
Team ID	23KD1A4233
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

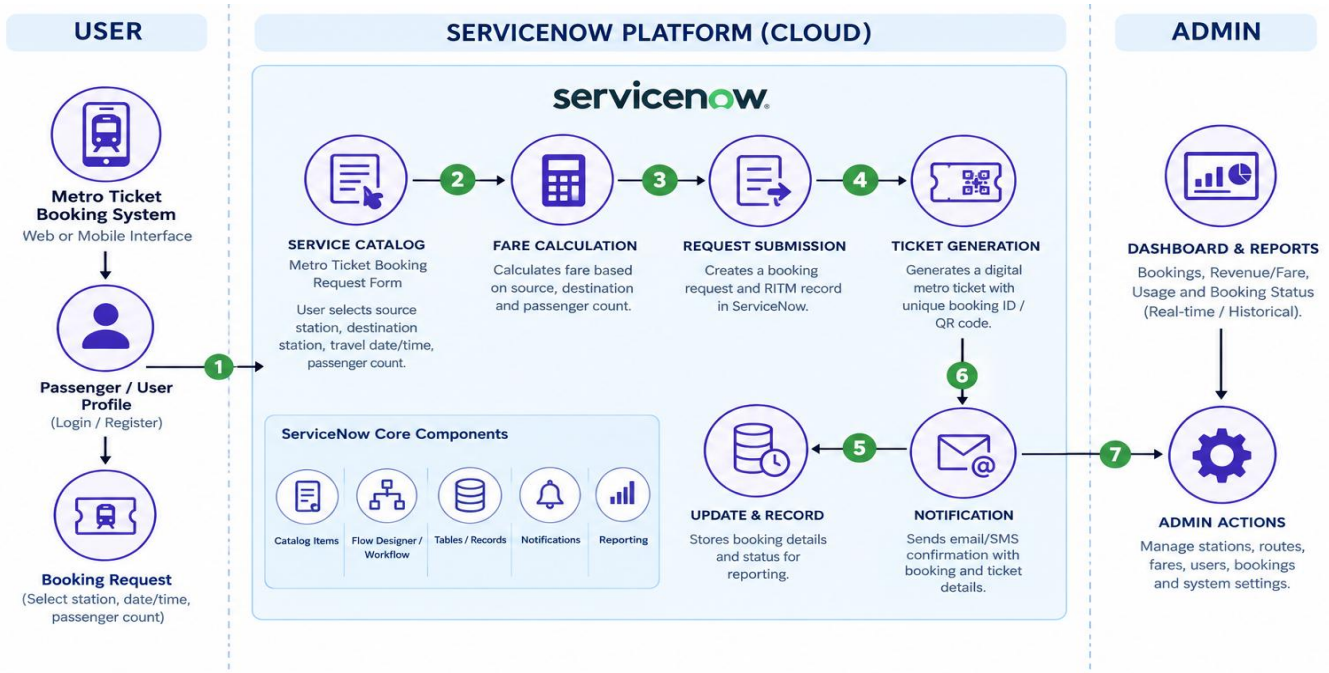


Table-1 : Components & Technologies:

S.No	Architecture Component	Core Responsibility & Function	Technology & Framework
1.	**Passenger Booking Portal**	Digital storefront for commuters to select routes, choose seats, and order tickets	ServiceNow Service Portal, Service Catalog, CSS3 / HTML5
2.	**Dynamic Fare Engine**	Client-side controller for dynamic route calculation based on station distance and passenger type	Catalog Client Scripts, Client Scripts (JavaScript)
3.	**Fare Validation Handler**	Asynchronous server-side processor for station validation and price verifications	Script Includes, GlideAjax API
4.	**Ticket Workflow Automator**	Orchestrates automated end-to-end booking approvals, ticket issue tasks, and dispatch	Flow Designer, Business Rules
5.	**Metro Operational DB**	Primary relational datastore holding station indexes, tariff schedules, and active ticket records	ServiceNow Tables (u_metro_station, u_metro_fare, sc_req_item)

6.	**QR Pass Generation Service**	Generates unique encrypted QR codes for gate access and digital verification	REST API / Scripted Third-Party QR Generator
7.	**Passenger Dispatch Notifier**	Real-time automated system sending digital tickets via email and SMS alerts	ServiceNow Event Management & Notification Engine
8.	**Predictive Analytics & Metrics**	AI-driven insights for passenger traffic trends, peak route usage, and revenue tracking	ServiceNow Predictive Intelligence, Performance Analytics & Dashboards
9.	**Digital Asset Repository**	Secure storage layer for generated QR code images, pass PDFs, and attachments	ServiceNow Attachment Management API
10.	**Platform Cloud Persistence**	Managed cloud infrastructure for scalable dataset persistence and high-availability backups	ServiceNow Cloud Database Engine
11.	**Enterprise SaaS Hosting**	High-availability cloud platform ensuring 24/7 runtime execution, isolation, and security	ServiceNow SaaS Cloud Platform (Multi-Tenant Infrastructure)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	**Client Scripting & Custom Logic**	Front-end dynamic form handling, real-time fare computation, and interactive field validations	JavaScript, GlideAjax API
2.	**Access Control & Data Security**	Granular role-based permissions (RBAC), end-to-end HTTPS encryption, and identity-driven platform governance	ServiceNow ACLs, User Roles, IAM Controls, HTTPS
3.	**Multi-Tenant Elasticity**	Distributed cloud infrastructure built to handle high peak-hour ticketing traffic and database expansions without manual scaling	ServiceNow Multi-Tenant Cloud Architecture, SaaS Infrastructure
4.	**System Resilience & Reliability**	Continuous service uptime backed by redundant cloud data centers, fault tolerance, and 24/7 automated platform monitoring	High-Availability Data Centers, ServiceNow Cloud Monitoring
5.	**Response Optimization**	Accelerated request processing via server-side asynchronous execution, script modularization	GlideAjax, Script Includes, Indexed Table Queries

References:

https://www.servicenow.com/docs/r/servicenow-platform/service-catalog/service-catalog.html

https://www.servicenow.com/docs/r/application-development/building-applications/flow-designer.html https://www.servicenow.com/docs/r/api-reference/c_GlideAjaxAPI.html

https://developer.servicenow.com